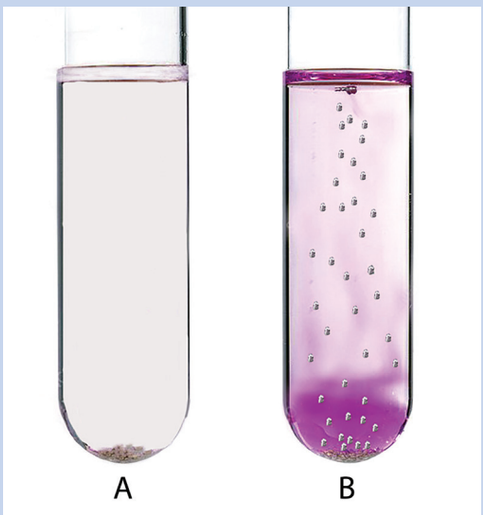
iv. Observe the two test tubes.

v. The observation shows that the solution in test tube B changes to pink colour and there is no such colour change in test tube A. That is, hot water reacts with magnesium according to the following reaction and there is no such reaction in cold water.

$$Mg + 2H_2O \longrightarrow Mg^{2+} + 2OH^- + H_2 \uparrow$$



vi. The resultant solution is basic and it is indicated by phenolphthalein.

A large number of reactions are known which do not take place at room temperature but occur readily at higher temperatures. Example: Reaction between H_2 and O_2 to form H_2O takes place only when an electric spark is passed.

Arrhenius suggested that the rates of most reactions vary with temperature in such a way that the rate constant is directly proportional to $e^{-\left(\frac{E_a}{RT}\right)}$ and he proposed a relation between the rate constant and temperature.

$$k = Ae^{-\left(\frac{E_a}{RT}\right)} \qquad ....(1)$$

Where A the frequency factor,

R the gas constant,

E_a the activation energy of the reaction and,

T the absolute temperature (in K)

The frequency factor (A) is related to the frequency of collisions (number of collisions per second) between the reactant molecules. The factor A does not vary significantly with temperature and hence it may be taken as a constant.

E_a is the activation energy of the reaction, which Arrhenius considered as the minimum energy that a molecule must have to posses to react.

Taking logarithm on both side of the equation (1)

$$\ln k = \ln A + \ln e^{-\left(\frac{E_a}{RT}\right)}$$

$$\ln k = \ln A - \left(\frac{E_a}{RT}\right) \qquad (\because \ln e = 1)$$

$$\ln k = \ln A - \left(\frac{E_a}{R}\right)\left(\frac{1}{T}\right) \qquad ....(2)$$

$$y = c + m\,x$$

The above equation is of the form of a straight line y = mx + c.

A plot of $\ln k$ Vs $\frac{1}{T}$ gives a straight line with a negative slope $-\frac{E_a}{R}$. If the rate constant for a reaction at two different temperatures is known, we can calculate the activation energy as follows.

At temperature $T = T_1$; the rate constant $k = k_1$

$$\ln k_1 = \ln A - \left(\frac{E_a}{RT_1}\right) \qquad ....(3)$$

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